

In no other major subdivision of business activity are to be found such great variations from one company to another between what goes in as expense and what comes out in benefits as occurs in research. Even among the best-managed companies this variation seems to run in a ratio of as much as two to one. By this is meant some well-run companies will get as much as twice the ultimate gain for each research dollar spent as will others. If averagely-run companies are included, this variation between the best and the mediocre is still greater. This is largely because the big strides in the way of new products and processes are no longer the work of a single genius. They come from teams of highly trained men, each with a different specialty. One may be a chemist, another a solid state physicist, a third a metallurgist and a fourth a mathematician. The degree of skill of each of these experts is only part of what is needed to produce outstanding results. It is also necessary to have leaders who can coordinate the work of people of such diverse backgrounds and keep them driving toward a common goal. Consequently, the number or prestige of research workers in one company may be overshadowed by the effectiveness with which they are being helped to work as a team in another.

Nor is a management's ability to coordinate diverse technical skills into a closely-knit team and to stimulate each expert on that team to his greatest productivity the only kind of complex coordination upon which optimum research results depend. Close and detailed coordination between research workers on each developmental project and those thoroughly familiar with both production and sales problems is almost as important. It is no simple task for management to bring about this close relationship between research, production, and sales. Yet unless this is done, new products as finally conceived frequently are either not designed to be manufactured as cheaply as possible, or, when designed, fail to have maximum sales appeal. Such research usually results in products vulnerable to more efficient competition.

Finally there is one other type of coordination necessary if research expenditures are to attain maximum efficiency. This is coordination with top management. It might perhaps better be called top management's understanding of the fundamental nature of commercial research. Development projects cannot be expanded in good years and sharply curtailed in poor ones without tremendously increasing the total cost of reaching the desired objective. The "crash" programs so loved by a few top managements may occasionally be necessary but are often just expensive. A